

radio hardware, which needs to be band-specific in order to receive and transmit at the required frequency. We expect to see innovation in this space as manufacturers work to solve this issue and produce programmable band selection in the hardware that will, in turn, allow devices to be produced that cover all the major bands worldwide," he adds.

"During the early stages of LTE growth, 3G HSPA or EV-DO will be the primary global roaming technologies given the global harmonisation of 3G spectrum. Over time, some common LTE bands across the world that facilitate international roaming will start to emerge. In fact, 3G and LTE will inter-work and coexist for a long time," says Dr Dondeti.

Facilitating international roaming is not a great problem, according to Dr Karandikar. "ITU has defined band plans for various technologies and standards. Various countries do harmonise their spectrum allocation accordingly. ITU holds World Radio Communications Conference every three to four years where frequency harmonisation may be achieved," he informs.

Global roaming issues have been solved in the past using multi-standard chipsets and multi-band radios. A similar solution might be helpful here too. Qualcomm has developed the world's first multi-mode (HSPA, EV-DO, FDD/TDD LTE) chipsets that integrate both TDD and FDD. The commercial LTE TDD network launched by Bharti Airtel uses multi-mode dongles based on Qualcomm's MDM 9x00.

"Huawei, ZTE, BandRich and Quanta have announced LTE TDD multi-mode devices based on the Qualcomm MDM9x00 chipset in August 2011, indicating commercial availability this year," adds Dr Dondeti.

When will 4G phase out 3G?

"Standards are quick to arrive and slow to leave. We have lived with 2G services for around 20 years now, for example. The question of when



Airtel 4G LTE USB modem

4G will phase out 3G should instead be phrased "when will 4G phase out 2G?," quips Maidment.

Today, in most territories, 2G and 3G services run side by side with 2G often covering less-densely populated regions and 3G services deployed in towns and cities. From a network perspective, we are seeing a growing trend towards heterogeneous network architectures which combine multiple carriers with varying access technologies—such as small cells to deliver capacity and coverage where it is needed most, and larger, more traditional macro cells providing wide-area coverage in rural areas.

As LTE is deployed both in the wide area through macro cells as well as in small cells, we expect to see operators looking to free up their existing 2G spectrum to move across to LTE services. This is a slow process which is highly dependent upon regulatory approval and will differ from region to region.

While some operators might consider leapfrogging directly from 2G to 4G, those who have deployed 3G are likely to continue enhancing and using those networks for at least five to ten years to get justifiable returns on their investments.

We can expect to see the continued adoption and rapid uptake of LTE services around the world. As a key part of that, we will see new low-power, multi-band capable devices that deliver the economies of scale required to make LTE commercially viable. From a standards perspective, device vendors are working to implement the first-generation LTE-Advanced terminals, which we would probably see a year or two down the line. These devices will be capable of support-

ing data rates up to 300 Mbps enabled by carrier aggregation and multi-layer multiple-input multiple-output.

"In addition to LTE-Advanced, there are plans within 3GPP to look at how LTE can be

employed to enable machine-type communications. The rise of machine-to-machine communications is anticipated to be a key growth area and it is essential that we have in place the right standards to support that need. Being an all-IP based technology, LTE is well positioned to serve this emerging space during the next 20+ years," says Maidment.

Beyond 4G

While 4G, according to its true definition, is not even here yet, we are already talking of what is yonder, beyond the big blue mountain.

Dr Borkar provides some futuristic food for thought: "Consistent with the general historical trend of a new technology standard every ten years, it is expected that '5G' specifications will likely be in place in the 2018-20 timeframe."

The framework for 5G includes higher-efficiency operation with lower battery consumption, higher system reliability, more uniform, high data rates across the coverage area, low infrastructure deployment costs, and higher spectral efficiency and capacity.

Some of the features being considered include advanced multi-cell coordination for enhanced system performance and end-user service quality, efficient infrastructure for machine-to-machine communications to support the vast number of connected devices, direct user device-to-device communications especially for performance and network outage situations, and new ways of using and adapting to the available spectrum based on the use of cognitive radios. ●

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